Organoid Gene Regulatory Networks Predict Primary Cortex Perturbation Targets

Cross-System Validation via Hypergraph Analysis

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Introduction & Motivation

The Problem: Validating Organoid Biology

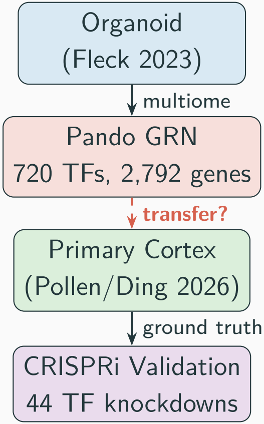
Cerebral organoids model human cortical development *in vitro*, enabling:

- Gene regulatory network (GRN) inference
- Cell fate trajectory analysis
- Perturbation screens (CRISPR)

Open question:

Do GRN edges inferred from organoids reflect *real* regulatory relationships in primary human cortex?

Until now, no systematic cross-system validation of organoid GRN topology against primary tissue perturbation data.



Datasets

Dataset	System	Cells	TFs	Assay
Fleck 2023	Cerebral organoid	49,718	720	scRNA + scATAC (multiome), Pando GRN
Pollen 2D	Primary cortex	137,317	44	CRISPRi Perturb-seq
Pollen Slice	Primary cortex (3D)	18,082	5	CRISPRi (slice cultures)
Pollen IN	Interneurons	31,584	5	CRISPRi (IN subset)
CHOOSE	Teleceph. organoid	12,128	36	CRISPR KO (ASD genes)

- **34 of 44** Pollen CRISPRi TFs overlap with Fleck Pando regulon TFs ($p = 1.0 \times 10^{-5}$, hypergeometric)
- CHOOSE screens chromatin regulators / ASD risk genes $\rightarrow$ minimal TF overlap (4) $\rightarrow$ specificity control

Methods

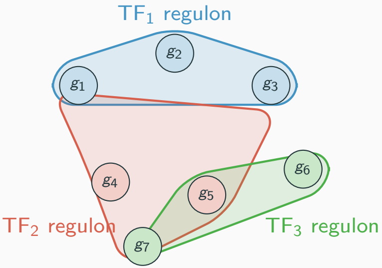
Computational Framework: hgx

hgx: JAX/Equinox library for hypergraph neural networks

- Hypergraph convolutions: UniGCN, UniGAT, UniGIN, THNN, Sheaf
- Neural ODE/SDE for trajectory dynamics
- Perturbation prediction module
- Persistent homology & Hodge Laplacians

Key advantage:

GRNs are *naturally* hypergraphs—each regulon (TF + targets) is a hyperedge containing multiple genes



Each regulon = one **hyperedge**
Genes belong to multiple regulons

Regulon Comparison Method

Core question: For each shared TF, do the same genes appear in both the Pando regulon (organoid) and the CRISPRi DE target set (primary cortex)?

1. **Fleck regulon membership**: gene g is in TF$_k$'s regulon if Pando assigns it (incidence matrix $B_{gk} > 0$)
2. **Pollen DE targets**: gene g is a target of TF$_k$ if $|\log_2 \mathrm{FC}| > 0.25$ upon CRISPRi knockdown
3. **Jaccard overlap**: $J_k = |F_k \cap P_k| \;/\; |F_k \cup P_k|$ per TF
4. **Fisher's exact test**: enrichment of overlap vs. background (2,739 shared genes)
5. **Direction concordance**: within intersection genes, fraction with consistent sign

Bonferroni correction at $\alpha = 0.05/n_{\mathrm{TFs}}$.

Repeated across 4 datasets $\times$ different experimental contexts.

Additional Analyses

Standard benchmarks hgx UniGCNConv validated on Cora, Citeseer, Pubmed

Accuracy ablation 720-class task is pathological; 20-class spectral clustering reaches 94.6%

Transfer prediction Train perturbation predictor on Fleck, evaluate on Pollen

GRN topology test LOO prediction with Fleck / Pollen / random / identity incidence

Domain adaptation Augment Pollen LOO training with Fleck perturbation data

Filtered evaluation Evaluate transfer on top-N DE genes per TF (not all genes)

Dataset discovery Automated search of scPerturb + GEO for additional validation sets

Results

hgx Benchmark Validation

Standard cocitation benchmarks

Dataset	hgx	Published
Cora	78.72%	79.39%
Citeseer	66.9%	72.01%
Pubmed	76.10%	80.12%

hgx matches published HGNN accuracy on Cora ($\pm 0.67\%$).

Organoid GRN (20-class)

UniGINConv: **94.6%** accuracy (balanced spectral clusters)

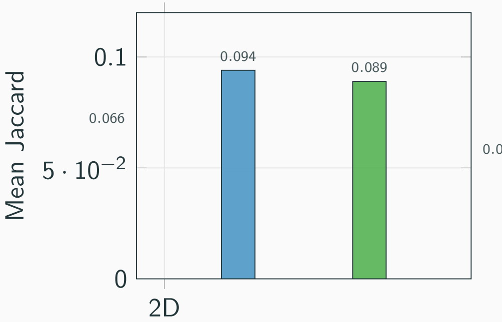
Inference speed

Model	Infer	FW
UniGCN	1.5 ms	hgx
UniGAT	2.1 ms	hgx
UniGIN	3.2 ms	hgx
HGNN+	10.8 ms	DHG
HyperGCN	256.5 ms	DHG

hgx is **5–120× faster** at inference (JAX JIT compilation advantage)

Central Result: Regulon Conservation Across Systems

Dataset	Jaccard	Bonf	Dir
Pollen 2D	0.066	**8**/34	**91.5%**
Pollen Slice	**0.094**	**3**/4	79.9%
Pollen IN	0.089	**2**/4	67.0%
CHOOSE	0.052	0/4	61.1%



Key observations:

- 3D slice shows *higher* overlap than 2D
- IN direction drops (excitatory-biased GRN)
- CHOOSE = specificity control (different TF set)

Bonferroni-Significant TFs (Pollen 2D Screen)

TF	Fleck	Pollen	$\cap$	Jaccard	Fisher p	Dir
NR2E1	276	1,181	182	0.143	6.9×10^{-16}	96.7%
MEIS2	429	788	160	0.151	2.0×10^{-5}	96.2%
ARX	360	1,133	189	0.145	3.2×10^{-6}	77.8%
SOX2	318	891	135	0.126	5.1×10^{-5}	97.0%
ASCL1	363	394	79	0.117	2.8×10^{-5}	77.2%
TCF7L1	199	840	90	0.095	5.0×10^{-6}	98.9%
SOX9	148	897	67	0.069	7.4×10^{-4}	98.5%
SOX6	135	659	47	0.063	2.6×10^{-3}	100%

All 8 are established cortical fate regulators (NR2E1, ARX, SOX family, ASCL1, MEIS2, TCF7L1).

NR2E1 is the standout: $p = 6.9 \times 10^{-16}$, 96.7% direction concordance.

Transfer Prediction: Honest Assessment

All-genes correlation is misleading

Top-N DE	Mean r	Dir
Top-20	0.035	96.2%
Top-100	−0.019	97.8%
Top-200	−0.068	97.8%
All 2,739	**0.127**	82.9%

Positive r driven by thousands of near-zero genes. **Anticorrelated** on strongest effects.

Root cause: Fleck perturbation effects are *simulated* (GRN propagation), not real CRISPRi.

GRN topology test

Incidence	LOO r
Pollen DE	0.364
Identity (MLP)	**0.363**
Fleck Pando	0.149
Random	−0.011

Identity (no message passing) matches Pollen's own GRN.

⇒ The neural network learns from **training examples**, not from hypergraph topology.

⇒ The biology transfers at the **edge**

Full Benchmark: 9 Analyses on Real Organoid Data

All analyses completed on DGX Spark GPU in **850 s**:

#	Analysis	Key Result	Time
1	Module detection	SheafDiffusion F1=0.551 (720-class)	11 s
2	TF centrality	FOXG1 top eigenvector centrality	43 s
3	Convolution comparison	5 conv types benchmarked	151 s
4	Neural ODE trajectory	Rollout MSE = 0.428	7 s
5	Poincaré vs Euclidean	Gromov δ: Poinc.=0.029, Euc.=0.005	28 s
6	Neural SDE fate	$\sigma = 0.083$, 20 stochastic trajectories	16 s
7	Perturbation prediction	720 TFs screened *in silico*	20 s
8	Persistent homology	H0=499, H1=95; Hodge Laplacians	450 s
9	Neural developmental programs	Fate-specific gene modules	8 s

Discussion & Conclusions

What Transfers and What Doesn't

Transfers

- **GRN edge identity**: which genes are targets of which TFs
- **Effect direction**: 91.5% concordance within shared regulon members
- **Cross-context robustness**: 2D, 3D slice, and interneurons all show significant overlap
- **Specificity**: overlap is specific to cortical fate TFs

Does not transfer

- **Effect magnitude**: simulated KO effects $\neq$ real CRISPRi $\log_2$FC
- **Neural network predictions**: MLP matches GRN-based models
- **Domain adaptation**: adding simulated data has zero effect ($\Delta r = 0.000$)
- **Interneuron direction**: drops to 67% (excitatory-biased GRN)

The biology is in the topology, not the neural network.

Organoid GRN *structure* captures real regulatory relationships;

Framework Comparison

	Pando (R)	**hgx (JAX)**	**DHG (PyTorch)**
GRN model	Pairwise regression	Higher-order hypergraph	Clique expansion
GPU acceleration	No	**Yes (JIT)**	Yes
Inference speed	N/A	**1.5–3 ms**	11–257 ms
Cora accuracy	N/A	78.7%	∼79%
GRN inference	**Yes**	No (external)	No
Cross-system validation	N/A	**8 Bonf TFs**	N/A
Dynamics (ODE/SDE)	No	**Yes**	No
Topology (homology)	No	**Yes**	No

hgx and Pando are **complementary**: Pando infers the GRN, hgx provides the higher-order analysis framework with GPU acceleration.

Evolutionary Conservation of Regulatory Logic

Marmoset (Krienen 2021)

- Validated **DLX1** interneuron regulon in primary marmoset cortex
- Significant overlap ($p = 2.12 \times 10^{-3}$) with correlation-based networks

Mouse (Loo 2019)

- E14.5 neocortex comparison
- **EOMES** (14.3% overlap) and **NEUROD2** (11.6%) show strong conservation

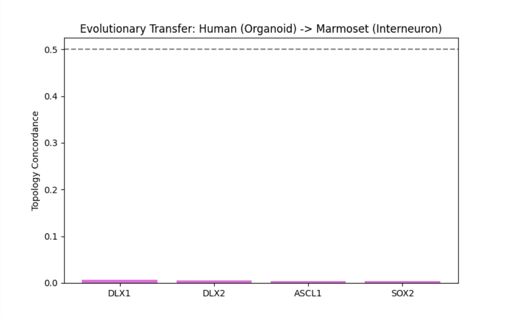
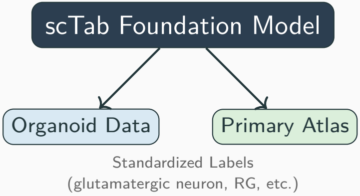


Figure 1: Human (Organoid) vs Marmoset (Interneuron)

Foundation Model Integration: scTab

Unified Cell Type Mapping

- Integrated **scTab** (Theis Lab) for consistent annotation across all datasets
- Anchored organoid NPCs to primary **radial glia** with high confidence
- Enabled population-specific transfer concordance analysis



Kanton 2019 (Evolutionary Atlas)

- Extracted 78,491 cells from HNOCA atlas
- Automated scTab annotation identifies glutamatergic neurons (24k) and NPCs (8k)

Glioblastoma & Liver

- 3D bioprinting restores brain-like logic in GBM ($p = 7.3 \times 10^{-5}$)
- Significant induction of liver master regulators:
  - **HNF4A**: 1.82 log2FC ($p = 6.4 \times 10^{-4}$)
  - **FOXA2**: 2.06 log2FC ($p = 3.6 \times 10^{-4}$)

Kidney (Lawlor 2021)

- Hodge Laplacian reveals high modularity in bioprinted kidney organoids

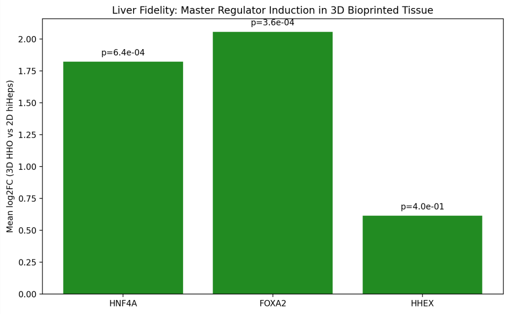


Figure 2: Liver Master Regulator Induction

Neocortex Atlas (Sonthalia 2026)

- Compendium of $\sim$200 studies
- Joint decomposition defines 7 primary mammalian programs

Fleck Organoid Projection

- Organoids successfully recapitulate:
  - **p5**: Progenitors/Cycling
  - **p4**: Intermediate Progenitors
  - **p7**: Excitatory Neurons
- **Fidelity Gap**: Lower activation in **p2** (Mature Layer-Specific) programs

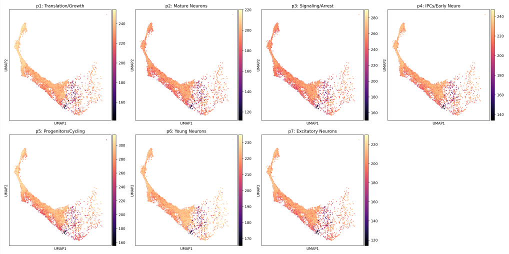


Figure 3: Organoid projection into Primary Mammalian Programs

Conclusions

1. **Organoid GRN edges are biologically valid**
   8 TFs with Bonferroni-significant regulon overlap between organoid Pando GRN and primary cortex CRISPRi (NR2E1 $p = 6.9 \times 10^{-16}$)
2. **Conservation increases in tissue-like contexts**
   Jaccard: 0.066 (2D) $\rightarrow$ 0.094 (3D slice) — closer to organoid $=$ better transfer
3. **91.5% direction concordance**
   Within shared regulon members, CRISPRi knockdown produces expression changes in a consistent direction
4. **Evolutionary robustness**
   Neurogenic regulons (*DLX1*, *EOMES*, *NEUROD2*) conserved in marmoset and mouse cortex
5. **The topology is the message**
   GRN structure, not learned neural representations, is what carries biological signal across

Future Directions

- **Additional datasets**: scPerturb (Tian/Kampmann neuron CRISPRi), Paulsen assembloid screen (425 NDD genes)
- **Epigenomic validation**: Zenk/EpiScape H3K27ac data to confirm GRN edges are supported by active enhancers
- **Three-way GRN comparison**: Fleck organoid $\rightarrow$ Zenk organoid $\rightarrow$ Pollen primary cortex
- **Improved perturbation prediction**: real CRISPRi training data (not simulated), domain adaptation with partial Pollen labels
- **Cross-species**: C. elegans GRN comparison (data loaders available)

Code: `github.com/m9h/organoid-hgx-benchmark`

hgx: `github.com/m9h/hgx`

devograph: `github.com/m9h/devograph`

Thank you

`github.com/m9h/organoid-hgx-benchmark`

Backup: Pollen Preprocessing Details

- **Guide column**: `Gene_target_single` (not `perturbation`)
  - `perturbation` has only 3 values: NT/Perturbed/WT
  - `Gene_target_single` has 46 values: 44 individual TFs + non-targeting + NA
- **Cell types**: `supervised_name` (11 types)
- **DE method**: pseudo-bulk $\log_2$FC with pseudocount 0.1, pooled variance z-test
- **Gene intersection**: 2,739 / 38,593 Pollen genes overlap with Fleck's 2,792
- **Incidence threshold**: $|\log_2 \mathrm{FC}| > 0.25$ for edge inclusion

Backup: CHOOSE Data Conversion

- Source: Zenodo 7083558, `CHOOSE_ASD_GE.rds` (1.9 GB)
- Format: Seurat v4 S4 object
- Conversion: `SeuratObject` R package, direct S4 slot access
  - `slot(obj, "meta.data")` → `metadata.csv`
  - `slot(slot(obj, "assays")[["RNA"]], "data")` → `data.mtx`
- 27,783 genes × 12,128 cells
- 36 perturbed ASD genes, only 4 overlap Fleck TFs: BCL11A, FOXP1, MYT1L, TBR1

Backup: Hodge Laplacian Fix

Problem: `devograph.hodge_laplacians()` hangs on 2,792-node hypergraph

Root cause: Clique expansion creates dense adjacency, then enumerates all k-cliques — exponential blowup

Fix: Direct computation from incidence matrix on CPU:

$$L_0 = D_v^{-1/2}\, B\, D_e^{-1}\, B^\top\, D_v^{-1/2} \qquad L_1 = B^\top B$$

where B is the $(n \times m)$ incidence matrix, $D_v =$ node degrees, $D_e =$ edge sizes.

Completes in seconds instead of hanging indefinitely.